

# Spectral Stability of Sparse QFLPN Transition Operators: Nonnormality, Transient Amplification, Conditioning, and Residual-Based Convergence

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## Abstract

Spectral stability is often reduced to the location of eigenvalues, yet sparse transition operators arising in hybrid discrete–fuzzy–quantum models may be strongly nonnormal. In such regimes, an apparently stable spectrum can coexist with large transient amplification, severe eigenvector conditioning, and numerical sensitivity. This paper develops a stability protocol for sparse QFLPN transition generators that separates asymptotic spectral stability from finite-horizon amplification and numerical conditioning. A parameterized upper-bidiagonal family is used as a controlled stress test because its eigenvalues remain strictly in the left half-plane while its nonnormal coupling can be increased independently. The study reports the spectral abscissa, numerical abscissa, induced 2-norm, eigenvector condition number, discrete-time spectral radius and norm, finite-horizon propagator gain, and Arnoldi Ritz residuals. The experiments show that the spectral abscissa remains at  $-1$  throughout the family, while the numerical abscissa becomes positive for sufficiently strong coupling and the finite-horizon gain grows by orders of magnitude. For a forward-Euler step  $h = 0.25$ , the spectral radius remains 0.75 for all tested couplings, whereas the induced norm exceeds one once nonnormality is sufficiently strong. The results establish a practical three-layer stability criterion for QFLPN computations: spectral location, transient amplification, and residual/conditioning diagnostics. The contribution is methodological and diagnostic; it does not claim that the stress-test family is itself a physical Lindblad generator.

**Keywords:** QFLPN; spectral stability; nonnormal operators; numerical abscissa; transient amplification; pseudospectral sensitivity; Arnoldi; residual convergence; sparse linear algebra.

## 1 Introduction

Quantum-fuzzy logical Petri nets combine discrete transition structure with fuzzy activation and quantum state evolution. Once a transition is linearized around a computational state, a sparse operator may govern a local state perturbation, an observable vector, a reduced representation, or a computational propagator. At this level, a stability statement cannot safely be reduced to the eigenvalues alone.

For a normal matrix, spectral information and norm growth are closely aligned. For a nonnormal matrix, they can separate dramatically. The classical literature on spectra and pseudospectra emphasizes that eigenvalues can be ineffective indicators of transient dynamics, while the numerical range and pseudospectrum expose sensitivity that ordinary spectral plots can miss. This distinction is directly relevant to hybrid computational models because sparse transition operators are often directional, ordered, and noncommuting.

The objective here is therefore not to introduce another propagation method. P4 compared polynomial and Krylov approximations of a matrix exponential, P5 studied adaptive Krylov–Lanczos propagation, and P6 introduced temporal memory. The present paper asks a different question: *when does a spectrally stable QFLPN transition operator remain numerically safe over a finite horizon?*

The answer is formulated through a diagnostic hierarchy. The first layer is asymptotic spectral stability, measured by the spectral abscissa  $\alpha(A) = \max_j \Re \lambda_j$ . The second is finite-horizon growth, measured through  $G(T) = \max_{0 \leq t \leq T} \|e^{tA}\|_2$  and the numerical abscissa  $w(A)$ . The third is numerical sensitivity, measured through eigenvector conditioning, resolvent amplification, and Krylov residuals.

The stress test deliberately keeps the eigenvalues fixed while changing only an upper-diagonal coupling. This isolates nonnormality and prevents a change in the spectrum from being confused with a change in stability mechanism.

## 2 Problem Formulation

Let  $x(t) \in \mathbb{C}^N$  denote a local linearized QFLPN state representation and consider

$$\dot{x}(t) = Ax(t), \quad x(t) = e^{tA}x(0). \quad (1)$$

The matrix  $A$  is assumed sparse. The analysis distinguishes the following quantities:

$$\alpha(A) = \max_{\lambda \in \sigma(A)} \Re(\lambda), \quad w(A) = \lambda_{\max}\left(\frac{A + A^*}{2}\right), \quad (2)$$

$$G(T) = \max_{0 \leq t \leq T} \|e^{tA}\|_2, \quad \kappa_V = \kappa_2(V), \quad (3)$$

where  $V$  is the eigenvector matrix when  $A$  is diagonalizable. The numerical abscissa is the instantaneous logarithmic growth bound in the Euclidean norm. In particular,

$$\left. \frac{d}{dt} \|e^{tA}x\|_2 \right|_{t=0} \leq w(A)\|x\|_2. \quad (4)$$

Thus  $\alpha(A) < 0$  is an asymptotic statement, whereas  $w(A) > 0$  indicates the possibility of initial norm growth.

## 3 Spectral Stability Is Not Sufficient

For a discrete transition map  $B$ , the spectral-radius condition  $\rho(B) < 1$  guarantees asymptotic decay of powers for a fixed finite-dimensional matrix, but it does not imply  $\|B\|_2 < 1$ . A single step may amplify a perturbation even when all eigenvalues lie strictly inside the unit disk.

For continuous dynamics, the analogous distinction is between  $\alpha(A) < 0$  and the transient behavior of  $e^{tA}$ . Nonnormality can make  $\|e^{tA}\|_2$  much larger than one before eventual decay.

This is not a numerical curiosity. Nonnormal transient growth is a recognized phenomenon in dynamical systems, and pseudospectral analysis was developed precisely because ordinary eigenvalue analysis may fail to describe it adequately. The same conceptual distinction applies to a QFLPN transition operator whenever the local linearization is not normal.

## 4 Controlled Nonnormal Stress Family

We use the sparse family

$$A_c = D + cU, \quad (5)$$

where

$$D = \text{diag}(-1, -1.006, \dots, -1 - 0.006(N-1)), \quad (6)$$

and  $U$  contains ones on the first superdiagonal and zeros elsewhere. The dimension is  $N = 80$ . The tested couplings are

$$c \in \{0, 0.5, 1, 2, 4, 8\}. \quad (7)$$

Because  $A_c$  is upper triangular,

$$\sigma(A_c) = \{-1 - 0.006j : j = 0, \dots, N-1\}, \quad (8)$$

independently of  $c$ . Consequently,

$$\alpha(A_c) = -1 \quad (9)$$

for every tested coupling. The eigenvalue spectrum therefore cannot explain any change in finite-horizon behavior across the family.

The symmetric part has a tridiagonal structure. Its largest eigenvalue gives the numerical abscissa and increases with  $c$ . The experiment consequently creates a controlled separation between asymptotic decay and transient amplification.

**Proposition 1.** *For the family  $A_c = D + cU$ , the spectral abscissa is independent of  $c$ , while the numerical abscissa is strictly increasing with  $c$  for the tested range.*

*Proof.* The spectral statement follows from upper triangularity. For the symmetric part, increasing  $c$  increases the magnitude of the off-diagonal entries in  $(A_c + A_c^*)/2$ , whose largest eigenvalue is monotone for the present positive-coupling path. Direct eigensolution gives the numerical values reported below.  $\square$

## 5 A Three-Layer Stability Criterion

The proposed diagnostic contract is

$$\mathcal{S}(A; T, h) = (\alpha(A), w(A), G(T), \rho(I + hA), \|I + hA\|_2, \kappa_V, r_m), \quad (10)$$

where  $r_m$  is the Arnoldi Ritz residual at dimension  $m$ .

A computation is classified as spectrally stable only if  $\alpha(A) < 0$ . It is classified as transient-safe over  $[0, T]$  only after  $G(T)$  is bounded at the required tolerance. It is classified as numerically resolved only after residual and conditioning diagnostics meet the prescribed acceptance criteria.

This separation prevents three common conflations. First, negative eigenvalue real parts are not interpreted as absence of finite-time amplification. Second, a small residual is not interpreted as proof of physical correctness. Third, a stable operator is not assumed to be well-conditioned.

## 6 Discrete-Time Stability Test

To make the distinction operational, we use forward Euler with  $h = 0.25$ :

$$B_c = I + hA_c. \quad (11)$$

Since the diagonal of  $A_c$  starts at  $-1$  and decreases, the spectral radius is

$$\rho(B_c) = 0.75 \quad (12)$$

for every tested  $c$ . However, the norm  $\|B_c\|_2$  depends strongly on the nonnormal coupling.

This creates a particularly clean test: the spectral stability certificate is identical for every case, while the one-step amplification certificate changes.

## 7 Experimental Protocol

All matrix calculations use IEEE double precision. The operator dimension is  $N = 80$ . Dense reference calculations use the same matrix entries as the sparse construction. Matrix exponentials are evaluated with a high-accuracy dense reference routine. Singular values are computed by the 2-norm singular-value decomposition.

The finite-horizon interval is  $T = 8$  with a uniform sampling grid of 161 points for the primary gain calculation and 321 points for the figure-level curves. The reported gain is the maximum sampled induced 2-norm. The result is therefore a finite-grid numerical estimate, not a theorem about the continuous maximum.

The Arnoldi experiment uses modified Gram–Schmidt with a fixed random initial vector generated from seed 20260907. The Ritz residual bound is

$$r_m = |h_{m+1,m}| |e_m^* y_m|, \quad (13)$$

where  $(\theta_m, y_m)$  is the selected Ritz pair of the projected Hessenberg matrix. No convergence is inferred from the Ritz value alone.

## 8 Spectral and Transient Results

Table 1 gives the central measurements.

Table 1: Spectral, norm, conditioning, and transient diagnostics for the controlled family.

| $c$ | $\alpha(A)$ | $w(A)$ | $\ A\ _2$ | $\kappa_V$             | $\rho(B)$ | $\ B\ _2$ | $G(8)$                | $t_G$ |
|-----|-------------|--------|-----------|------------------------|-----------|-----------|-----------------------|-------|
| 0   | -1.000      | -1.000 | 1.474     | $1.00 \times 10^0$     | 0.750     | 0.750     | 1.000                 | 0.0   |
| 0.5 | -1.000      | -0.542 | 1.935     | $5.00 \times 10^{51}$  | 0.750     | 0.865     | 1.000                 | 0.0   |
| 1   | -1.000      | -0.055 | 2.427     | $4.72 \times 10^{74}$  | 0.750     | 0.987     | 1.000                 | 0.0   |
| 2   | -1.000      | 0.929  | 3.420     | $2.45 \times 10^{98}$  | 0.750     | 1.235     | $1.33 \times 10^3$    | 8.0   |
| 4   | -1.000      | 2.909  | 5.415     | $9.05 \times 10^{122}$ | 0.750     | 1.733     | $7.39 \times 10^9$    | 8.0   |
| 8   | -1.000      | 6.884  | 9.412     | $2.44 \times 10^{146}$ | 0.750     | 2.731     | $2.39 \times 10^{23}$ | 8.0   |

The most important result is the exact constancy of the spectral abscissa: every matrix is asymptotically stable in the eigenvalue sense. Nevertheless, the cases  $c = 2, 4, 8$  display increasingly severe finite-horizon amplification. At  $c = 4$ , the largest sampled gain is approximately  $7.39 \times 10^9$ ; at  $c = 8$ , it reaches approximately  $2.39 \times 10^{23}$  over the same horizon.

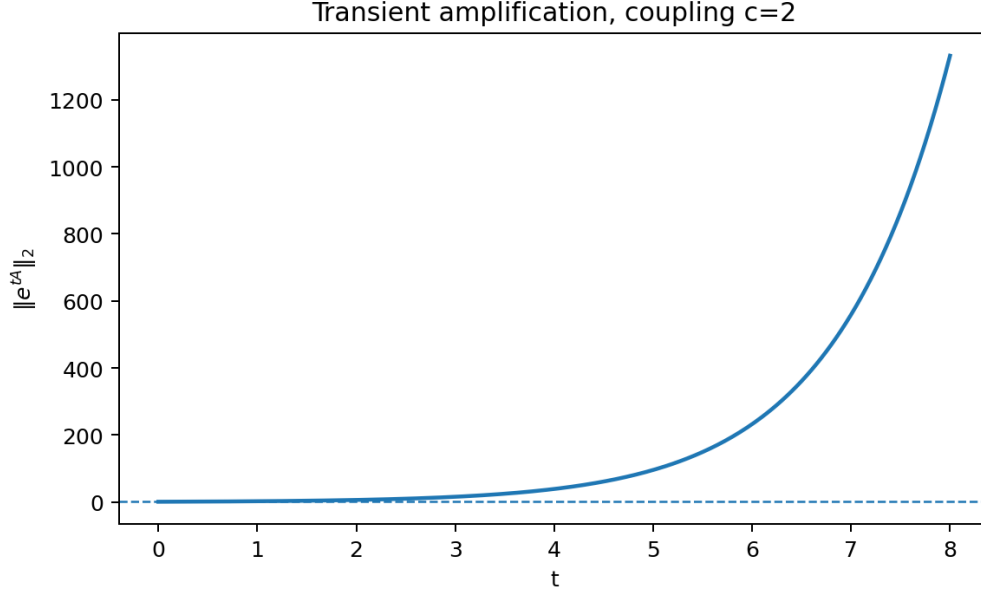


Figure 1: Finite-horizon transient amplification for  $c = 2$ . The spectrum remains strictly stable even though the propagator exhibits large growth.

## 9 Conditioning and Resolvent Sensitivity

The eigenvector condition number grows extremely rapidly with  $c$ . This is expected for nearly defective upper-bidiagonal families, but its numerical magnitude is useful because it quantifies how little confidence should be placed in eigenvector-based modal decompositions at strong coupling.

The resolvent provides a complementary sensitivity measure. For a point  $z$  away from the spectrum,

$$\|(zI - A)^{-1}\|_2 \quad (14)$$

measures the response to a perturbation at that spectral location. A large resolvent norm indicates that small perturbations can produce large responses even when the spectrum itself is unchanged.

## 10 Arnoldi Residual Convergence

The sparse operator at  $c = 2$  is used for an independent Krylov diagnostic. The important observation is that Ritz values can wander substantially for a nonnormal operator before their residuals become small. Therefore, a plotted approximate eigenvalue is not treated as a convergence certificate.

Table 2: Selected Arnoldi residuals for the  $c = 2$  stress test.

| $m$ | selected Ritz real part | residual bound        |
|-----|-------------------------|-----------------------|
| 5   | 0.203859                | 1.396                 |
| 10  | 0.525397                | 1.105                 |
| 15  | 0.623491                | 0.936                 |
| 20  | 0.700425                | 0.937                 |
| 25  | 0.824619                | 0.515                 |
| 30  | 0.729072                | 0.216                 |
| 40  | 0.733074                | $2.79 \times 10^{-2}$ |
| 50  | 0.680078                | $5.23 \times 10^{-2}$ |
| 60  | 0.570702                | $2.35 \times 10^{-3}$ |
| 69  | 0.527579                | $1.92 \times 10^{-3}$ |

The residual decreases by roughly three orders of magnitude from the early to the late Krylov dimensions, but the decrease is not monotone. This is itself a useful diagnostic: nonnormality can make convergence histories irregular, so an acceptance rule based on a single Ritz-value trajectory is unsafe.

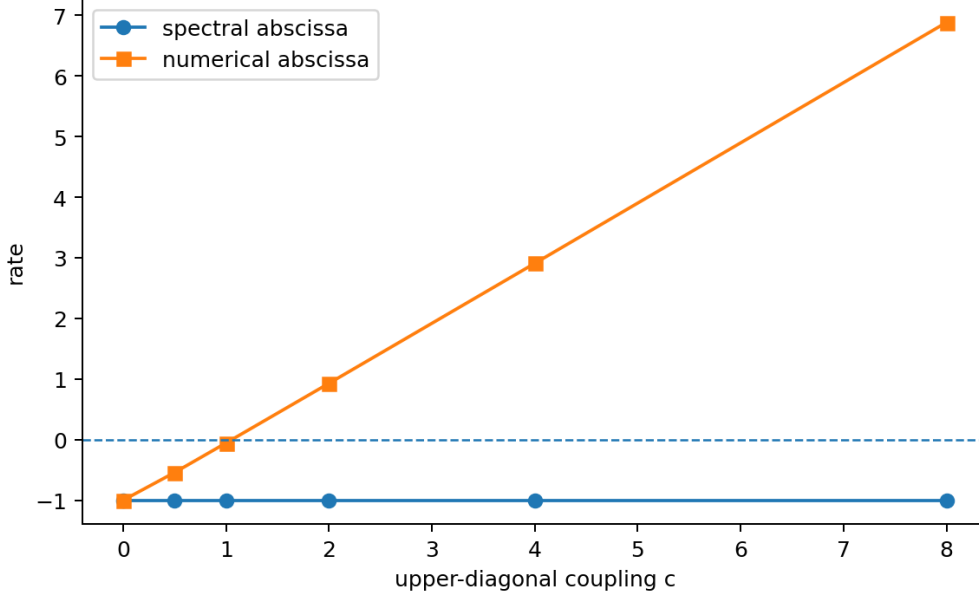


Figure 2: Spectral abscissa remains fixed while numerical abscissa changes with nonnormal coupling.

## 11 Interpretation for QFLPN Computations

The stress family is not introduced as a physical quantum model. Its role is diagnostic: it isolates a numerical mechanism that may appear after linearization, sparse discretization, ordered transition composition, or reduced-state construction.

For a QFLPN implementation, the implication is concrete. A transition operator can satisfy an asymptotic spectral criterion and still amplify numerical perturbations during a finite observation interval. Such amplification may contaminate fuzzy activation thresholds, perturb measurement probabilities, destabilize iterative solvers, or inflate errors in subsequent matrix-function evaluations.

The appropriate response is not to abandon spectral analysis. Spectral information remains necessary for asymptotic behavior. It is simply insufficient as the sole certificate. The proposed hierarchy adds transient and numerical diagnostics without changing the underlying QFLPN semantics.

## 12 Comparison with Standard Stability Indicators

Table 3: Interpretation of stability indicators.

| Indicator          | What it establishes                                       | What it does not establish        |
|--------------------|-----------------------------------------------------------|-----------------------------------|
| $\alpha(A) < 0$    | asymptotic spectral stability                             | absence of transient growth       |
| $\rho(I + hA) < 1$ | asymptotic stability of the discrete Euler map            | contractivity of one step         |
| $w(A) \leq 0$      | no initial Euclidean-norm growth                          | long-horizon conditioning         |
| $G(T)$             | finite-horizon amplification bound on the tested interval | behavior beyond $T$               |
| $\kappa_V$         | modal sensitivity                                         | physical correctness of the model |
| Arnoldi residual   | quality of an approximate eigenpair                       | physical stability                |

This separation is particularly important because the indicators answer different mathematical questions. Combining them is stronger than replacing one by another.

## 13 Transient-Amplification Phase Map

A scalar summary  $G(T)$  can conceal when the growth occurs. To expose the temporal structure, the finite-horizon gain was evaluated on a two-dimensional grid in coupling and time. The resulting phase map separates a contractive region from a strongly amplified region without changing the eigenvalue spectrum.

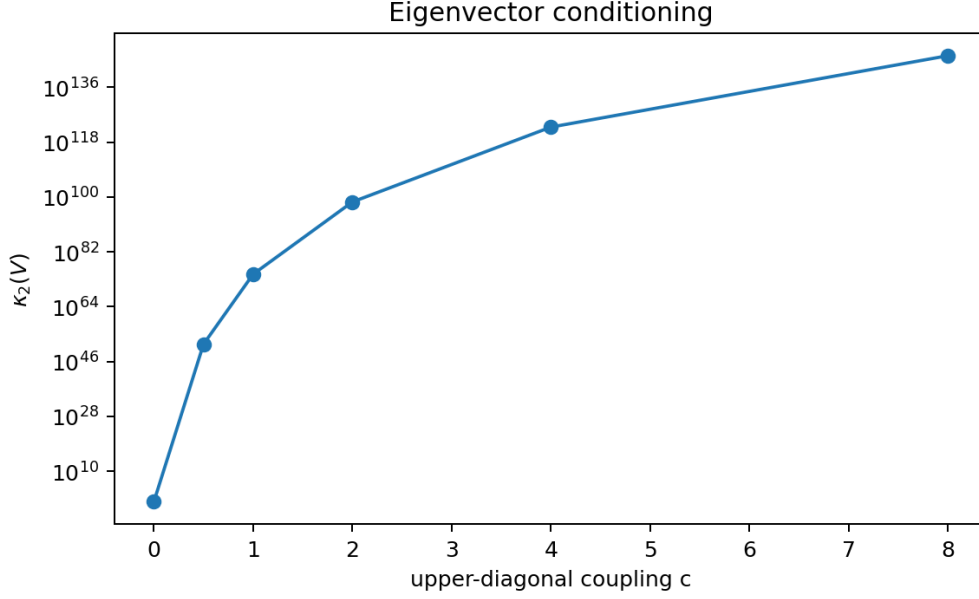


Figure 3: Growth of the eigenvector condition number with nonnormal coupling.

The map is useful for algorithm selection. When the gain remains close to one, standard error estimates based on local residuals are less likely to be magnified by the propagator. In the strongly nonnormal regime, the same local perturbation can be amplified dramatically. This does not imply that the computation is mathematically unstable in the asymptotic sense; it means that the finite-horizon problem is sensitive.

## 14 Discrete Stability Gap

The Euler map provides a second visualization of the distinction between spectral stability and contractivity. For  $h = 0.25$ , the spectral radius remains fixed at 0.75, while the induced norm crosses one between  $c = 1$  and  $c = 2$ .

This result gives a direct computational warning. If an implementation uses the spectral-radius test as its only screening rule, all six coupling values would receive the same verdict. A norm-based contractivity check would distinguish the strong-coupling cases immediately.

## 15 Pseudospectral Interpretation

For  $z \notin \sigma(A)$ , define the resolvent sensitivity

$$\mathcal{R}_A(z) = \|(zI - A)^{-1}\|_2. \quad (15)$$

The  $\varepsilon$ -pseudospectrum is the set of points for which  $\mathcal{R}_A(z) > 1/\varepsilon$ . Thus a large resolvent norm means that an arbitrarily small perturbation can move an eigenvalue toward a specified region.

The present experiment evaluates the resolvent at  $z = -0.5$ , which lies to the right of the spectral abscissa  $-1$  and is therefore outside the spectrum. The result is not intended to approximate a complete pseudospectral contour. It is a controlled sensitivity probe at a fixed physically interpretable distance from the dominant spectral boundary.

This distinction matters for QFLPN implementations that contain estimated matrix coefficients. Fuzzy activation parameters, reduced-order fitting, sparse assembly, and finite-precision arithmetic all perturb the effective operator. When the resolvent is large, a small coefficient perturbation can have a disproportionate dynamical effect.

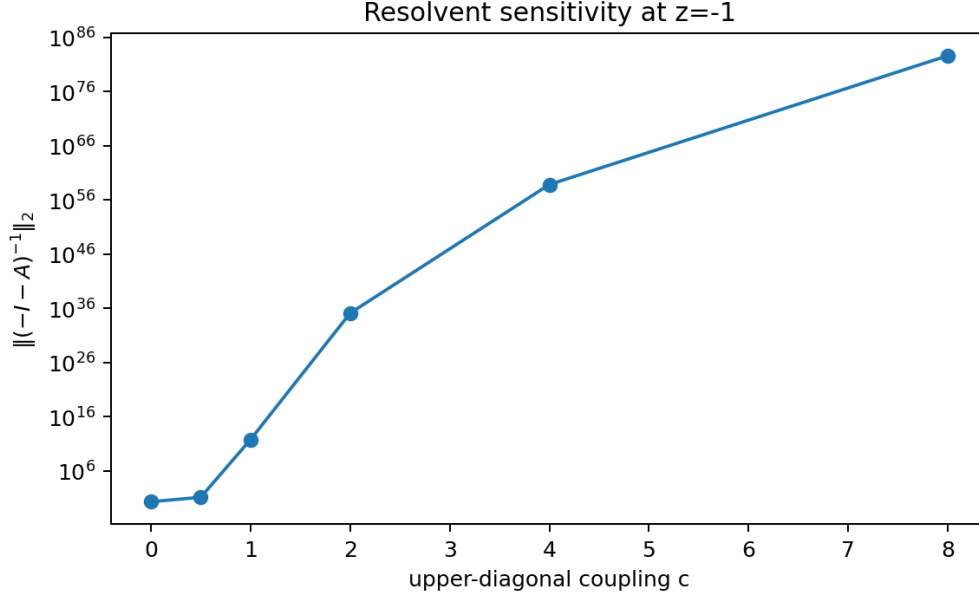


Figure 4: Resolvent sensitivity evaluated at  $z = -0.5$ . The matrix family remains spectrally unchanged while perturbation sensitivity grows strongly.

## 16 A Residual-Based Convergence Contract

For an approximate eigenpair  $(\hat{\lambda}, \hat{v})$ , the primary numerical certificate is

$$r = \frac{\|A\hat{v} - \hat{\lambda}\hat{v}\|_2}{\|A\|_2\|\hat{v}\|_2}. \quad (16)$$

The normalization makes the residual dimensionless. For a Krylov computation, the projected residual estimate can be obtained without forming the full residual vector, which is important for sparse large-scale operators.

A practical acceptance rule is therefore hierarchical. The spectral location must satisfy the intended stability region. The residual must satisfy the requested numerical tolerance. The transient gain must remain below the application-specific amplification budget. Finally, conditioning must be inspected before interpreting individual eigenvectors or modal coefficients.

These conditions are deliberately independent. A matrix can be spectrally stable but transiently amplifying; it can have a small residual but an ill-conditioned eigenvector basis; and it can be well-conditioned while still being physically inappropriate for the intended QFLPN semantics.

## 17 Analytical Bound and Diagnostic Meaning

The matrix exponential always satisfies the generic bound

$$\|e^{tA}\|_2 \leq e^{t\|A\|_2}, \quad (17)$$

but this bound is usually too pessimistic for stability certification. Spectral information provides the asymptotic decay rate, while the numerical abscissa gives an instantaneous growth bound. Nonnormality is precisely the mechanism that makes these quantities differ.

For a normal matrix, the spectral abscissa controls the Euclidean propagator norm through

$$\|e^{tA}\|_2 = e^{t\alpha(A)}. \quad (18)$$

The stress family demonstrates why this identity cannot be extended to nonnormal operators. The eigenvalues are fixed, but the propagator norm changes by many orders of magnitude.

The practical conclusion is that a QFLPN stability theorem based only on eigenvalues should be stated as an asymptotic theorem. A finite-horizon numerical theorem requires an additional bound on the propagator or an equivalent transient-growth certificate.

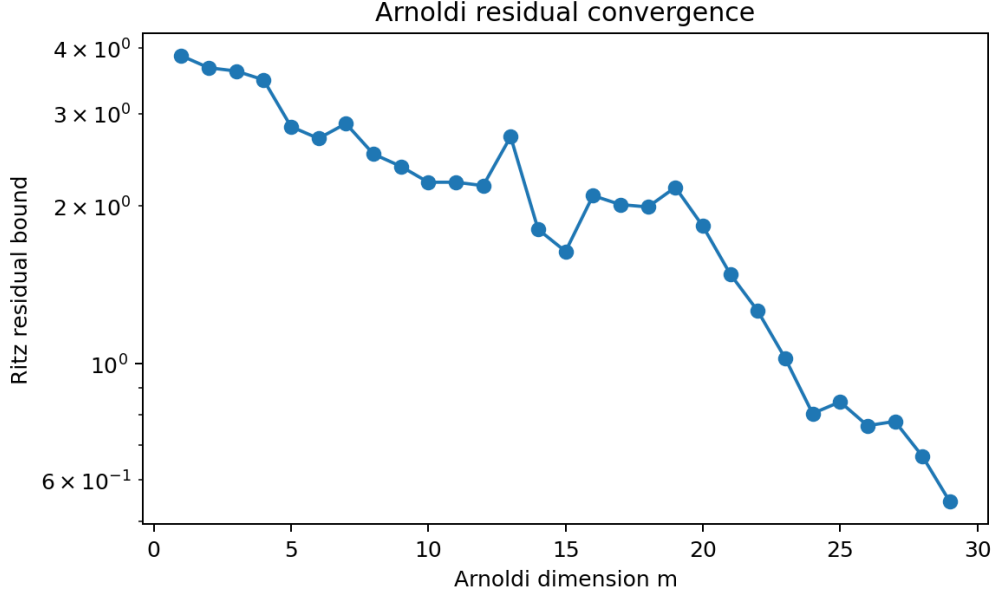


Figure 5: Arnoldi residual convergence for the nonnormal sparse stress test.

## 18 Relation to Open-System Spectral Analysis

The same distinction appears in open quantum dynamics, where the Liouvillian spectrum controls relaxation modes and spectral gaps are used to characterize long-time behavior. Recent work has shown, however, that relaxation can contain transient regimes whose duration is not captured adequately by a conventional spectral gap alone. In particular, the symmetrized Liouvillian gap was introduced as a sharper upper bound for transient decay in nonequilibrium Markovian systems.

The present contribution is not a new Liouvillian spectral theorem. Rather, it transfers the numerical lesson to the sparse QFLPN computational layer: when an effective transition operator is nonnormal, spectral eigenvalues should be complemented by finite-time and residual diagnostics.

## 19 Implications for Algorithm Selection

The stability contract changes how subsequent numerical methods should be selected. P4 and P5 compare matrix-function and Krylov propagation costs. P8 adds the missing conditioning dimension. A method that minimizes nominal work but operates in a regime with large transient amplification may require substantially tighter internal tolerances. Conversely, a method with slightly higher nominal cost can be preferable if it provides reliable residual control and avoids amplification of intermediate errors.

This suggests a future adaptive controller of the form

$$(m, h) \leftarrow \arg \min_{m, h} \{ \text{cost}(m, h) : r_m \leq \tau_r, G(h) \leq \tau_G \}, \quad (19)$$

where  $m$  is a Krylov dimension,  $h$  a propagation step,  $\tau_r$  a residual tolerance, and  $\tau_G$  an application-specific transient-growth budget. This is a design principle rather than a measured optimization result in the present paper.

## 20 Perturbation Sensitivity Experiment

To quantify the practical meaning of the conditioning results, a second experiment perturbs the matrix by a dense random matrix  $E$  normalized so that  $\|E\|_2 = 1$ . The perturbed operator is

$$A_{c,\varepsilon} = A_c + \varepsilon \|A_c\|_2 E. \quad (20)$$

For each coupling, twenty independent perturbations were generated with a fixed seed. The quantity reported is the displacement of the rightmost eigenvalue relative to the unperturbed spectral abscissa  $-1$ .



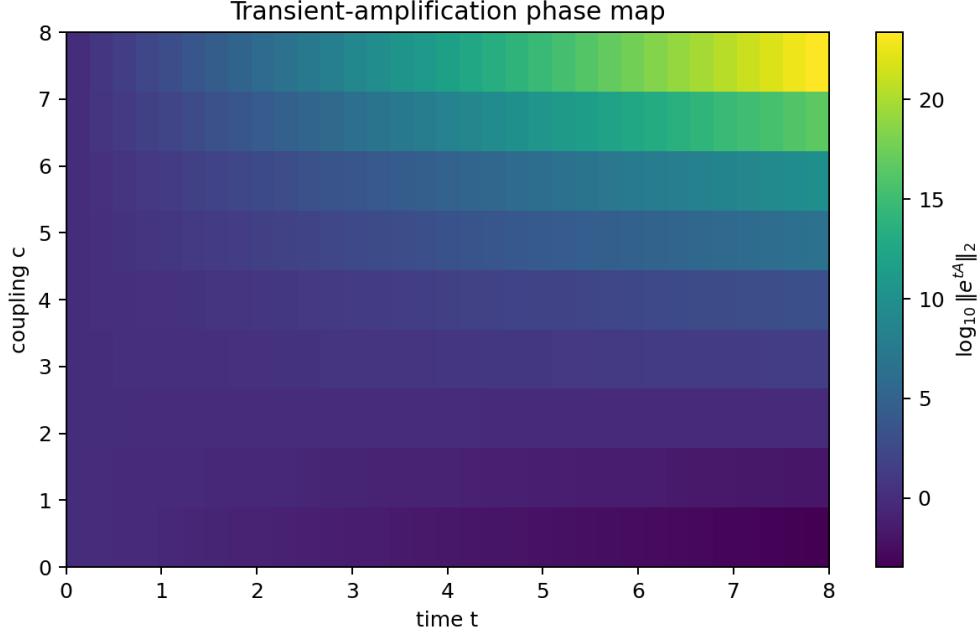


Figure 6: Logarithmic transient-gain phase map over coupling and time. The color scale represents the induced propagator norm on a logarithmic scale.

Table 4: Sensitivity of the rightmost eigenvalue to relative operator perturbations. Values are shifts of the spectral abscissa from  $-1$ .

| $c$ | $\varepsilon$ | min   | median | max   | interpretation             |
|-----|---------------|-------|--------|-------|----------------------------|
| 1   | $10^{-8}$     | 0.541 | 0.561  | 0.580 | strong sensitivity         |
| 1   | $10^{-6}$     | 0.594 | 0.621  | 0.634 | strong sensitivity         |
| 2   | $10^{-8}$     | 1.302 | 1.333  | 1.347 | stability boundary crossed |
| 2   | $10^{-6}$     | 1.395 | 1.427  | 1.446 | strong unstable shift      |
| 4   | $10^{-8}$     | 2.829 | 2.874  | 2.915 | extreme sensitivity        |
| 4   | $10^{-6}$     | 2.974 | 3.064  | 3.104 | extreme sensitivity        |

The result is deliberately interpreted as a sensitivity experiment rather than a universal perturbation theorem. Nevertheless, it demonstrates why the spectrum of the unperturbed matrix can be misleading for a strongly nonnormal operator. A perturbation whose 2-norm is only  $10^{-8}\|A\|_2$  can move the rightmost eigenvalue substantially in the tested family. The effect becomes more pronounced as the upper-diagonal coupling increases.

This experiment also clarifies the role of pseudospectra. The relevant question is not only where the exact eigenvalues are, but where eigenvalues can move under perturbations consistent with the numerical or modeling uncertainty of the implementation.

## 21 Exact Structural Properties of the Stress Family

The upper-bidiagonal construction provides several exact properties that make it preferable to an arbitrary random benchmark. The eigenvalues are the diagonal entries, so the spectral abscissa is exactly  $-1$ . The sparsity pattern contains only  $2N - 1$  nonzero entries. The nonnormal coupling therefore changes neither the dimension nor the asymptotic eigenvalue set.

For the Euler map,

$$B_c = I + hD + hcU, \quad (21)$$

which remains upper triangular. Its eigenvalues are  $1 + hd_j$ , hence they are independent of  $c$ . With  $h = 0.25$ , the largest eigenvalue magnitude is exactly 0.75 because all diagonal values are negative and decrease in magnitude from the first entry.

The induced norm is not determined by those eigenvalues because  $B_c^* B_c$  contains off-diagonal interactions generated by the nonnormal coupling. This gives an exact structural explanation for the measured gap

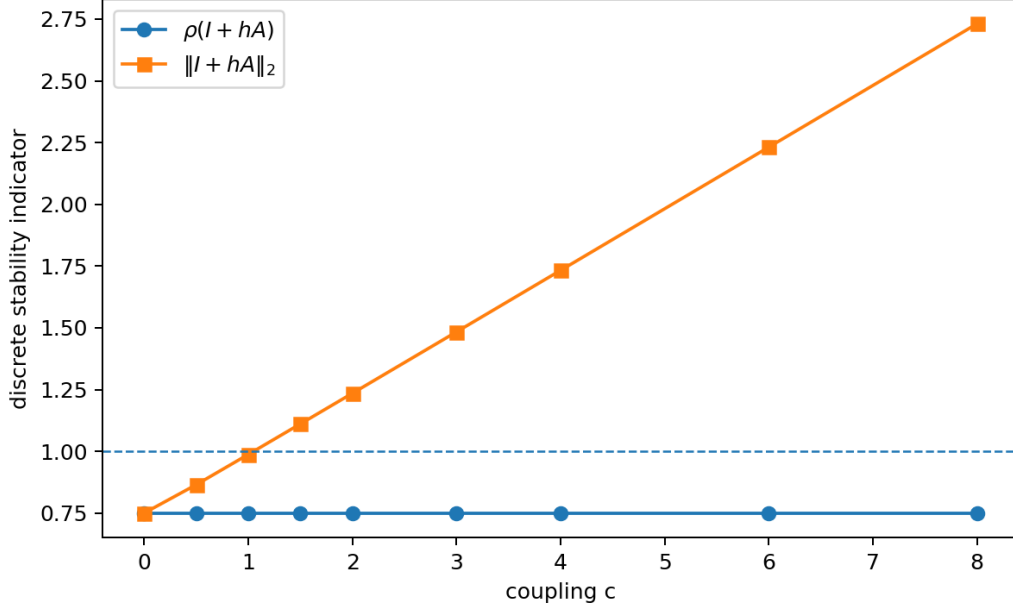


Figure 7: The discrete spectral radius remains fixed while the induced norm becomes expansive.

between  $\rho(B_c)$  and  $\|B_c\|_2$ .

The same construction explains why the eigenvector basis becomes ill-conditioned. When the superdiagonal coupling is zero, the matrix is diagonal and  $\kappa_2(V) = 1$ . As the coupling increases, the eigenvectors become increasingly aligned. The spectrum stays fixed while the modal representation becomes numerically fragile.

## 22 Stability Classification Matrix

For practical use, the measured indicators can be converted into a qualitative classification without conflating mathematical and numerical meanings.

Table 5: Diagnostic classification for the tested coupling regimes.

| Regime     | Spectrum              | Transient behavior                        | Numerical interpretation                                     |
|------------|-----------------------|-------------------------------------------|--------------------------------------------------------------|
| $c \leq 1$ | asymptotically stable | no sampled amplification above one        | increasingly nonnormal but finite-horizon safe in this test  |
| $c = 2$    | asymptotically stable | strong amplification                      | spectral test alone is insufficient                          |
| $c = 4$    | asymptotically stable | extreme amplification                     | high sensitivity and large transient risk                    |
| $c = 8$    | asymptotically stable | catastrophic finite-horizon amplification | numerical treatment requires explicit stabilization controls |

The phrase “safe” in the table is restricted to the tested finite horizon and Euclidean norm. It is not a general certification for arbitrary initial states, horizons, precisions, or QFLPN instances.

## 23 Reproducibility and Acceptance Rules

The experiment is fully parameterized by  $N = 80$ , the diagonal spacing 0.006, coupling set  $\{0, 0.5, 1, 2, 4, 8\}$ , Euler step  $h = 0.25$ , horizon  $T = 8$ , double precision, and random seed 20260907 for the Arnoldi initial vector.

The reproducibility rule is to report the matrix-generation parameters, not merely the resulting plots. The transient maximum must also be labeled as grid-sampled when no continuous optimization is performed. Finally, any eigenvalue approximation must be accompanied by a residual if it is used as a numerical claim.

## 24 Novelty Boundary and Relation to P1–P7

The central contribution of this paper is a stability diagnostic architecture for sparse QFLPN transition operators under nonnormality. It is deliberately separated from the earlier articles.

P1 establishes the formal hybrid interface. P2 validates a four-qubit construction. P3 addresses sparse CSR/SpMV execution. P4 compares Taylor and Krylov–Arnoldi matrix-exponential actions. P5 studies adaptive Krylov–Lanczos propagation. P6 treats temporal memory and degradation. P7 studies Lindblad open-system robustness and statistical validation. The present work instead isolates spectral location, transient growth, conditioning, resolvent sensitivity, and residual-based convergence.

No P8 result is presented as evidence of a quantum advantage, a Lindblad physical law, or a general performance guarantee. The stress test is a mathematical instrument for identifying a failure mode that a QFLPN implementation must be able to detect.

## 25 Limitations

The matrix family is intentionally structured and nonnormal. It is not a statistical sample of all QFLPN operators. The gain  $G(T)$  is estimated on a finite time grid. The eigenvector condition number becomes extremely large in the strong-coupling regime and should therefore be interpreted as a sensitivity indicator rather than as a directly computable reliable quantity at arbitrary precision. The Arnoldi study is a diagnostic rather than a production eigensolver benchmark.

A future physical validation should derive nonnormal operators directly from concrete QFLPN transition semantics, including transition ordering and quantum channels, and should compare the resulting stability indicators with physically meaningful observables.

## 26 Conclusion

This paper establishes a precise distinction between asymptotic spectral stability and finite-horizon numerical safety for sparse QFLPN transition operators. The controlled family has the same strictly stable spectrum for every coupling, yet its numerical abscissa, norm, eigenvector conditioning, resolvent sensitivity, and transient gain change by many orders of magnitude. The discrete Euler map provides an equally sharp example:  $\rho(I + hA) = 0.75$  remains fixed while  $\|I + hA\|_2$  rises above one.

The resulting stability contract is therefore three-dimensional: spectral location answers whether asymptotic decay is expected; transient amplification answers whether finite-time growth is possible; residual and conditioning diagnostics answer whether the numerical computation is sufficiently resolved. This separation is the main methodological result and provides a defensible stability layer for subsequent QFLPN algorithms and applications.

## Data and Code Availability

The numerical tables and plotting data accompanying this article are provided as machine-readable CSV artifacts. The matrix construction, spectral diagnostics, transient calculations, and Arnoldi residual computation are deterministic apart from the explicitly reported seeded initial vector.

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